

Mathematics for Data Science

Session 1 - Vector spaces, subspaces, and linear maps

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Definition

Let G be a non-empty set. G is a group if there exists an operation $\star$ such that:

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Let G be a non-empty set. G is a group if there exists an operation $\star$ such that:

- i) **Closure.** For any $x, y \in G$, $x \star y$ also belongs to G .
- ii) **Associativity.** For any $x, y, z \in G$, $(x \star y) \star z = x \star (y \star z)$.
- iii) **Neutral element/Identity.** There exists $e \in G$ such that $x \star e = e \star x = x$ for all $x \in G$.
- iv) **Symmetric element/Inverse.** For each $x \in G$, there exists an element $x' \in G$, such that $x \star x' = x' \star x = e$.

Commutative group

Definition

Let G be a group. If for any $x, y \in G$, $x \star y = y \star x$ then we say that the group is commutative.

NB: “Abelian” group is another name for commutative groups

Vector spaces

Vector spaces: the basic setting in which linear algebra happens
Elements of V are called **vectors**.

¹A field is an algebraic structure that is closed under the four usual arithmetic operations $+$; $-$; $\times$; $\div$.

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Vector space

A vector space $(V, +, \cdot)$ over $\mathbb{R}$ is a set endowed with two operations:

- $+: V \times V \rightarrow V$ that allows to sum two vectors: $(x, y) \mapsto x + y$
- $\cdot: \mathbb{R} \times V \rightarrow V$ that is the multiplication of a vector by a **scalar**:
 $(a, x) \mapsto a \cdot x$

Vector spaces can be defined over any **field** $\mathbb{F}$. We take $\mathbb{F} = \mathbb{R}$ in this course. ¹

¹A field is an algebraic structure that is closed under the four usual arithmetic operations $+$; $-$; $\times$; $\div$.

Vector spaces

$(V, +, \cdot)$ must follow a number of axioms:

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$(V, +, \cdot)$ must follow a number of axioms:

- $(V, +)$ is a commutative group.
 - **Associativity.** For any $x, y, z \in V$, $(x + y) + z = x + (y + z)$
 - **Additive identity.** There exists an element in V , denoted $\mathbf{0}$, such that $x + \mathbf{0} = x$ for all $x \in V$.
 - **Additive inverse.** For any $x \in V$, there exists an element in V , denoted $-x$, such that $x + (-x) = \mathbf{0}$.
 - **Commutativity.** For any $x, y \in V$, $x + y = y + x$.

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 - **Commutativity.** For any $x, y \in V$, $x + y = y + x$.
- The law $\cdot$ verifies:
 - **Distributivity over scalar addition.** For all $x \in V$ and $\alpha, \beta \in \mathbb{R}$, $(\alpha + \beta)x = \alpha x + \beta x$.
 - **Distributivity over vector addition.** For all $x, y \in V$ and $\alpha \in \mathbb{R}$, $\alpha(x + y) = \alpha x + \alpha y$.
 - **“Mixed” associativity.** For all $x \in V$ and $\alpha, \beta \in \mathbb{R}$, $\alpha(\beta x) = (\alpha\beta)x$.
 - **Multiplicative identity.** There exists an element in $\mathbb{R}$, denoted 1 , such that $1 \cdot x = x$ for all $x \in V$.

Remark

Can we subtract vectors ?

Remark

Can we subtract vectors ?

We can subtract vectors because a subtraction is the addition of the opposite vector.

Examples of Vector Spaces

Example

- $(\mathbb{R}, +, \cdot)$ is a vector space where $\mathbb{R}$ is the set of real numbers
- $\mathbb{R}$ is also a vector space over $\mathbb{Q}$, with the usual addition and scalar multiplication by rational numbers.
- $\mathbb{C}$ is a vector space over $\mathbb{R}$, and also over $\mathbb{Q}$.
- $(\mathbb{R}^n, +, \cdot)$ is a vector space, called the **Euclidean space**.

ML illustration: data points as vectors

In many machine learning problems, each data point is represented as a vector.

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{pmatrix} \in \mathbb{R}^d$$

- x_i may be a feature: age, income, pixel intensity, word count, etc.
- A dataset with m observations can be viewed as a collection of vectors $x^{(1)}, \dots, x^{(m)} \in \mathbb{R}^d$.

ML illustration: images as vectors

A grayscale image of size 28×28 can be represented as a vector in $\mathbb{R}^{784}$.



Figure 1: The MNIST dataset contains 70,000 images of handwritten digits (0 to 9) that have been size-normalized and centered in a square grid of pixels. Each image is a $28 \times 28 \times 1$ array of floating-point numbers representing grayscale intensities ranging from 0 (black) to 1 (white).

ML illustration: images as vectors

In the case of the MNIST dataset, the target data consists of one-hot binary vectors of size 10, corresponding to the digit classification categories zero through nine.

$$x = \begin{pmatrix} \text{pixel}_1 \\ \text{pixel}_2 \\ \vdots \\ \text{pixel}_{784} \end{pmatrix}; \quad y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_{10} \end{pmatrix}$$

Thus, image classification can be seen as learning a function

$$f : \mathbb{R}^{784} \rightarrow \{0, \dots, 9\}.$$

NB: in practice, the function f often maps the input to a vector of probabilities over the 10 classes, and the predicted class is the one with the highest probability.

Examples of Vector Spaces

Example

- The set of matrices with $n \times p$ real-valued coefficients $\mathcal{M}_{n,p}(\mathbb{R})$
- The set of functions from an interval $I \subset \mathbb{R}$ to $\mathbb{R}$
- The set of sequences in $\mathbb{R}$
- The set of real polynomials of degree at most n ,

$$\mathbb{R}_n[X] = \left\{ \sum_{i=0}^n a_i X^i \mid a_0, \dots, a_n \in \mathbb{R} \right\},$$

with the usual addition and scalar multiplication:

$$(P + Q)(X) = \sum_{i=0}^n (a_i + b_i) X^i, \quad (\lambda P)(X) = \sum_{i=0}^n \lambda a_i X^i.$$

Product of vector spaces

Definition

Cartesian product of two sets A and B , denoted $A \times B$:

$$A \times B = \{(a, b), a \in A, b \in B\}$$

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Definition

Let E and F be two $\mathbb{F}$ -vector spaces. The sum of $(x, y) \in E \times F$ and $(x', y') \in E \times F$ is defined as $(x, y) + (x', y') = (x + x', y + y')$. The multiplication of (x, y) by a scalar $\lambda \in \mathbb{F}$ is defined as $\lambda \cdot (x, y) = (\lambda \cdot x, \lambda \cdot y)$.

Product of vector spaces

Proposition

Endowed with the above operations, $(E \times F, +, \cdot)$ is a $\mathbb{F}$ -vector space. It is called the **product vector space** of E by F .

Proposition

Let $n > 1$ an integer, we suppose that $E_1, \dots, E_n$ are n finite-dimensional vector spaces over $\mathbb{F}$. Then their cartesian product is finite-dimensional, with dimension

$$\dim(E_1 \times E_2 \times \cdots \times E_n) = \sum_{i=1}^n \dim(E_i).$$

Exercise: Proof (reasoning by recurrence)

Example: Euclidean space (again !)

Let n be a positive integer. We saw that $(\mathbb{R}^n, +, \cdot)$ is a vector space, called the **Euclidean space**.

We can see $\mathbb{R}^n$ as the product of n copies of $\mathbb{R}$:

$$\mathbb{R}^n = \underbrace{\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}}_{n \text{ times}}.$$

The vectors in the Euclidean space consist of n -tuples of real numbers, i.e.

for $x \in \mathbb{R}^n$, $x = (x_1, x_2, \dots, x_n)$. The x_i are the components or entries of the vector.

NB: you can think of vectors of $\mathbb{R}^n$ as $n \times 1$ matrices, or **column vectors**.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

ML illustration: product vector spaces and multimodal data

Suppose an object is described by two types of features:

$$u \in \mathbb{R}^{d_1} \quad \text{image features}, \quad v \in \mathbb{R}^{d_2} \quad \text{text features}.$$

The combined representation is

$$(u, v) \in \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}.$$

Equivalently, we can concatenate the features:

$$(u, v) \longleftrightarrow \begin{pmatrix} u \\ v \end{pmatrix} \in \mathbb{R}^{d_1+d_2}.$$

Examples of Operations

Example

- Summation of two vectors in $\mathbb{R}^2$.
- Multiplication by a scalar in $\mathbb{R}^n$.

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- Multiplication by a scalar in $\mathbb{R}^n$.

Addition and scalar multiplication are defined component-wise on vectors in $\mathbb{R}^n$:

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}, \quad \alpha \mathbf{x} = \begin{bmatrix} \alpha x_1 \\ \vdots \\ \alpha x_n \end{bmatrix}$$

ML illustration: mean image

In the MNIST dataset, we can compute the mean image $\bar{x}_c$ of each class c by averaging the pixel values across all images in the training set for this class.

$$\bar{x}_c = \frac{1}{m_c} \sum_{i=1}^{m_c} x^{(i)} \in \mathbb{R}^{784}$$

where $x^{(i)}$ is the vector representation of the i -th image in the training set for class c , and m_c is the total number of images in the training set for class c .

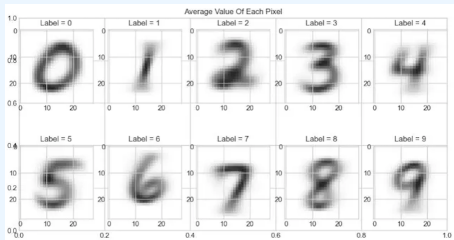


Figure 2: Mean image for each class of the MNIST dataset.

ML illustration: A simple template matching approach for classification

For a new image $x \in \mathbb{R}^{784}$, we can compute the correlation with the mean image $\bar{x}_c$ as follows:

$$\text{corr}(\bar{x}_c, x) = \frac{\sum_{j=1}^{784} (\bar{x}_{c,j} - \bar{\bar{x}})(x_j - \bar{x})}{\sqrt{\sum_{j=1}^{784} (\bar{x}_{c,j} - \bar{\bar{x}})^2} \sqrt{\sum_{j=1}^{784} (x_j - \bar{x})^2}}$$

where $\bar{\bar{x}}$ is the mean of the pixel values in the mean image, and $\bar{x}$ is the mean of the pixel values in the new image.

Finally, the predicted class of the new image can be determined by finding the class of the mean image that has the highest correlation with the new image:

$$\hat{y} = \arg \max_{c \in \{0,1,\dots,9\}} \text{corr}(\bar{x}_c, x)$$

where $\bar{x}_c$ is the mean image for class c .

ML illustration: average feature vector

Let $x^{(1)}, \dots, x^{(m)} \in \mathbb{R}^d$ be data points.

Their empirical mean is

$$\bar{x} = \frac{1}{m} \sum_{i=1}^m x^{(i)}.$$

This is used, for example, when centering data:

$$x^{(i)} \mapsto x^{(i)} - \bar{x}.$$

Centering is a basic preprocessing step in many ML algorithms.

Geometrical interpretation

In $\mathbb{R}^n$, a vector $\mathbf{x}$ has a **direction** and a **length** (or **norm**).

Scalar multiplication changes the length but keeps the same direction (i.e., on the same line). If the scalar is negative, the orientation is reversed.

Parallel vectors: We say that two vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ are **parallel** if there is a non-zero scalar $r \in \mathbb{R}$ such that

$$\mathbf{w} = r\mathbf{v}.$$

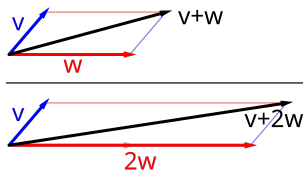


Figure 3: Vector addition and scalar multiplication: a vector $\mathbf{v}$ (blue) is added to another vector $\mathbf{w}$ (red, upper illustration). Below, $\mathbf{w}$ is multiplied by a factor of 2, yielding the sum $\mathbf{v} + 2\mathbf{w}$. (image from Wikipedia)

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Subspaces

Vector spaces can contain other vector spaces.

Subspaces

If V is a vector space, then $S \subseteq V$ is said to be a **subspace** of V if

- i) $\mathbf{0} \in S$
- ii) S is closed under addition: $\mathbf{x}, \mathbf{y} \in S$ implies $\mathbf{x} + \mathbf{y} \in S$
- iii) S is closed under scalar multiplication: $\mathbf{x} \in S, \alpha \in \mathbb{R}$ implies $\alpha\mathbf{x} \in S$

Subspaces

Example

- $\{0\}$ is always a subspace of V .
- The trivial vector space which contains only 0 .
- A line passing through the origin is a subspace of the Euclidean space.
- The set of real-valued sequences that converge is a subspace of the set of real-valued sequences.
- The set of continuous functions from an interval $I \subset \mathbb{R}$ into $\mathbb{R}$ is a subspace of the set of functions from I to $\mathbb{R}$.
- The set of polynomials of degree at most $k < n$ is a subspace of the set of polynomials of degree at most n .

Subspaces

Proposition

Let V be a vector space over $\mathbb{F}$. Any subspace U of V is a vector space over $\mathbb{F}$.

Exercise: Proof. By definition, the null element belongs to U . Verify that the composition rules $+$, $\cdot$ defined on V also hold for U .

Linear Combination

Linear combination

Given vectors $v_1, v_2, \dots, v_k \in \mathbb{R}^n$ and scalars $c_1, c_2, \dots, c_k \in \mathbb{R}$ we say that a vector of the form

$$w = c_1 v_1 + c_2 v_2 + \dots + c_k v_k$$

is a **linear combination** of the vectors $v_1, v_2, \dots, v_k$ with scalar coefficients $c_1, c_2, \dots, c_k$.

NB : The scalars c_i are sometimes called **weights**.

Span (*Sous-espace vectoriel engendré*)

Span

The **span** of a set of vectors $v_1, v_2, \dots, v_k \in \mathbb{R}^n$ is the set of all the vectors which can be written as a linear combination of the $v_1, v_2, \dots, v_k$, i.e.

$$\text{span}\{v_1, v_2, \dots, v_k\} = \{c_1 v_1 + c_2 v_2 + \dots + c_k v_k \mid c_1, c_2, \dots, c_k \in \mathbb{R}\}.$$

NB: equivalent to the French notation $\text{Vect}(v_1, v_2, \dots, v_k)$

Example

- $\text{span}((1, 0), (0, 1)) = \mathbb{R}^2$
- What is the span of $((2, 0), (0, 1))$?
- What is the span of $((1, 0), (3, 0))$?

ML illustration: span and representation

Suppose we have vectors $v_1, \dots, v_k \in \mathbb{R}^d$.
Their span contains all vectors of the form

$$c_1 v_1 + \dots + c_k v_k.$$

In ML, this can be interpreted as representing a data point using a set of basic patterns:

$$x \approx c_1 v_1 + \dots + c_k v_k.$$

This idea appears in dictionary learning, dimensionality reduction, and representation learning.

ML illustration: span and dictionary learning

In **dictionary learning**, we try to represent data vectors as linear combinations of a few basic vectors called **atoms**.

$$d_1, \dots, d_k \in \mathbb{R}^n, \quad D = \begin{bmatrix} | & & | \\ d_1 & \cdots & d_k \\ | & & | \end{bmatrix} \in \mathbb{R}^{n \times k}.$$

A data point $x \in \mathbb{R}^n$ is approximated by

$$x \approx c_1 d_1 + \cdots + c_k d_k = Dc, \quad c = (c_1, \dots, c_k)^\top.$$

Thus, dictionary learning searches for vectors $d_1, \dots, d_k$ such that many data points lie close to

$$\text{span}\{d_1, \dots, d_k\}.$$

Interpretation: the dictionary atoms are elementary patterns; the coefficients say how to combine them.

Properties of subspaces

Proposition

Let F and G be subspaces of a vector space E . Then $F \cap G$ is a subspace of E .

Proposition

Let $n \in \mathbb{N}$, $a_1, a_2, \dots, a_n \in \mathbb{R}$. The following set F is a vector subspace of $\mathbb{R}^n$.

$$F = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n \mid \sum_{i=1}^n a_i x_i = 0\}$$

Geometric interpretation

Given p_0 , fixed in $\mathbb{R}^n$, a line in $\mathbb{R}^n$ is given by (cf affine equation in $\mathbb{R}^2$):

$$\ell = \{x \in \mathbb{R}^n \mid x = p_0 + tv, t \in \mathbb{R}\}.$$

A special case is $\text{span}\{v\} = \{tv \mid t \in \mathbb{R}\}$, the line through the origin ($p_0 = 0$) in direction v .

Example

Let $p_1 = (1, 2)$ and $p_2 = (3, 1)$ be on

$$\ell = \{(1, 2) + (2, -1)t \mid t \in \mathbb{R}\}.$$

Is $p_1 + p_2$ on the line ?

Geometric interpretation

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Is $p_1 + p_2$ on the line ?

NB : If $u, w \in \text{span}\{v_1, \dots, v_k\}$, then $u + w \in \text{span}\{v_1, \dots, v_k\}$.
However, the sum of two points on a line need not be on the line.

Geometric interpretation

A plane in $\mathbb{R}^n$ through p_0 with directions $\mathbf{u}, \mathbf{v}$ is

$$P = \{x \in \mathbb{R}^n \mid x = p_0 + s\mathbf{u} + t\mathbf{v}, s, t \in \mathbb{R}\}.$$

If $p_0 = 0$, then $P = \text{span}\{\mathbf{u}, \mathbf{v}\}$.

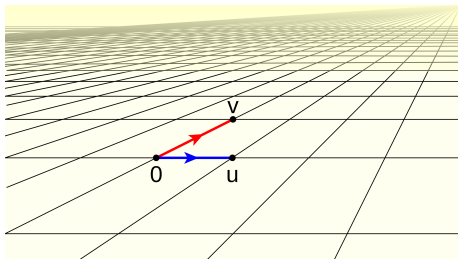


Figure 4: The cross-hatched plane is the linear span of $\mathbf{u}$ and $\mathbf{v}$ in both $\mathbb{R}^2$ and $\mathbb{R}^3$ (figure in perspective from Wikipedia).

Geometric interpretation

Affine subspaces are obtained by translating a vector subspace by a fixed vector p_0 .

: In $\mathbb{R}^n$, the affine subspace through p_0 with directions $v_1, \dots, v_k$ is:

$$P = \{x \in \mathbb{R}^n \mid x = p_0 + s_1 v_1 + \dots + s_k v_k, s_1, \dots, s_k \in \mathbb{R}\}.$$

If $p_0 = 0$, then $P = \text{span}\{v_1, \dots, v_k\}$.

NB: If the vectors $v_1, \dots, v_k$ are linearly independent, this affine subspace has dimension k .

Spanning List

Spanning list

A **spanning list** of V is a list whose span is V .

NB: French speakers “famille génératrice de V ”

The family of vectors $v_1, v_2, \dots, v_k$ spans the $\mathbb{R}$ -vector space V if for any $x \in V$, there exist coefficients $\lambda_1, \lambda_2, \dots, \lambda_k$ such that $x = \sum_{i=1}^k \lambda_i v_i$.

Example

- Verify that $\{(2, 1), (0, 1)\}$ is a spanning list of $\mathbb{R}^2$.
- Let us consider $V = \{(x, 0, 0), x \in \mathbb{R}\}$ and $v_1 = (1, 0, 0)$. Verify that V is a vector space and that (v_1) is a spanning list of V .

Linear Independence

Linear independence

A list/family of vectors $v_1, \dots, v_k$ is **linearly independent**/free if the only linear combination that equals the zero vector is the trivial one, i.e. where all coefficients are zero,

$$\alpha_1 v_1 + \dots + \alpha_k v_k = 0 \quad \Rightarrow \quad \alpha_1 = \dots = \alpha_k = 0$$

Equivalently, no vector in the family can be written as a linear combination of the others.

Example

Are the following vectors linearly independent ?

$$v_1 = (1, 0, 0), v_2 = (3, 0, 0), v_3 = (0, 0, 1)$$

$$v_1 = (1, 0), v_2 = (0, 2)$$

ML illustration: linear dependence and redundant features

Consider a tabular dataset with three features:

x_1 = height in meters, x_2 = height in centimeters, x_3 = weight.

Since $x_2 = 100x_1$, the first two features are linearly dependent. Then the second feature does not bring a new independent direction compared to the first. They are redundant.

Some ML algorithms, such as linear regression, can be sensitive to redundant features. ²

²The mathematical reason is that the design matrix may become singular, which can lead to numerical instability and poor generalization. NB: We call the design matrix the matrix whose rows are the data points and whose columns are the features. The design matrix X is singular if its columns are linearly dependent, i.e. if there exists a non-zero vector c such that $Xc = 0$. Thus, it is often useful to identify and remove linearly dependent features.

Basis of a vector space

Basis

A linearly-independent spanning list of vectors from a vector space V is called a **basis** of V .

Basis of a vector space

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Example

The standard basis of $\mathbb{R}^n$, called the **canonical basis**:

$$e_1 = (1, 0, \dots, 0), \quad e_2 = (0, 1, 0, \dots, 0), \quad \dots, \quad e_n = (0, \dots, 0, 1)$$

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Proposition

Given a basis, every vector has a unique coordinate representation.

ML illustration: one-hot encoding and bases

A categorical variable with k possible categories can be represented using the standard basis of $\mathbb{R}^k$:

$$e_1 = (1, 0, \dots, 0), \quad e_2 = (0, 1, 0, \dots, 0), \quad \dots, \quad e_k = (0, \dots, 0, 1).$$

This is called **one-hot encoding**.

Each category is represented by one basis vector.

ML illustration: embeddings

One-hot vectors are sparse, and often high-dimensional. For example: a dictionary with k words.

Instead, ML models often learn lower-dimensional dense vectors called **embeddings**:

$$\text{word} \mapsto x \in \mathbb{R}^d.$$

For example:

$$\text{cat, dog, car} \mapsto x_{\text{cat}}, x_{\text{dog}}, x_{\text{car}} \in \mathbb{R}^d.$$

Embeddings represent objects as vectors whose geometry can encode useful information.

ML illustration: image embeddings

An image I can be mapped by a neural network Φ to a vector

$$\Phi(I) = z \in \mathbb{R}^d.$$

$$\text{image} \longmapsto \text{embedding vector } z = \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_d \end{pmatrix}.$$

Goal: images with similar semantic content should have nearby embeddings.

$$\text{similar images} \Rightarrow \text{nearby vectors in } \mathbb{R}^d.$$

Examples:

- images of the same object should have close embeddings;
- images from different categories should be farther apart.

ML illustration: image retrieval with embeddings

Suppose each image I is represented by an embedding

$$z = \Phi(I) \in \mathbb{R}^d.$$

To find images similar to a query image I_q , compute

$$z_q = \Phi(I_q),$$

then search for images whose embeddings are close to z_q .

For example, using Euclidean distance (NB: see Lecture on norms):

$$\|z_q - z_i\|_2.$$

Interpretation: image retrieval becomes a nearest-neighbor problem in a vector space.

Dimension of a vector space

Proposition

Let V be a finite-dimensional vector space. Then:

- 1 V admits a basis.
- 2 All bases of V have the same cardinality.
- 3 Any family of linearly independent vectors from V can be completed to obtain a basis of V (FR: “*théorème de la base incomplète*”)
- 4 A basis of V can be extracted from any spanning family of V (FR: “*théorème de la base extraite*”).

Dimension

All bases of a vector space have the same length, called the **dimension**.

Example

$$\dim(\mathbb{R}^n) = n$$

Rank of a family of vectors

Definition

Let $\mathcal{F} = (v_1, v_2, \dots, v_k)$ be a family of vectors from a vector space V . The **rank** of $\mathcal{F}$ is the dimension of its span, i.e. the dimension of the vector space generated by all the linear combinations of vectors from $\mathcal{F}$.

$$\text{rank}(\mathcal{F}) = \dim(\text{span}(\mathcal{F}))$$

ML illustration: rank and redundant tabular features

Consider a tabular dataset with three features:

x_1 = height in meters, x_2 = height in centimeters, x_3 = weight.

Since

$$x_2 = 100x_1,$$

the first two features are linearly dependent.

For a data matrix

$$X = \begin{bmatrix} | & | & | \\ x_1 & x_2 & x_3 \\ | & | & | \end{bmatrix} \in \mathbb{R}^{m \times 3},$$

the columns x_1 and x_2 span only one direction.

$$\text{rank}(X) < 3.$$

Interpretation: rank reveals linear redundancy among features.

ML illustration: rank and face images

A grayscale face image with p pixels can be represented as a vector

$$x \in \mathbb{R}^p.$$

A dataset of m face images can be stored as a matrix

$$X = \begin{bmatrix} - & - & - & x^{(1)\top} & - & - & - \\ - & - & - & x^{(2)\top} & - & - & - \\ & & & \vdots & & & \\ - & - & - & x^{(m)\top} & - & - & - \end{bmatrix} \in \mathbb{R}^{m \times p}.$$

Although p may be very large, face images often vary mainly along a smaller number of directions: lighting, pose, expression, identity.

$$\text{rank}(X) \ll \min(m, p) \quad \text{or approximately so.}$$

Interpretation: the dataset has low-dimensional structure inside a high-dimensional pixel space.

Sum of subspaces

Definition

Let F and G be subspaces of E . Their sum is defined as

$$F + G = \{x \in E \mid \exists u \in F, w \in G, x = u + w\}$$

Exercise: Verify that this set is also a subspace of E .

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Definition

If $F \cap G = \{\mathbf{0}\}$, the sum is said to be a **direct sum** and written $F \oplus G$.

NB : (*sous-espaces vectoriels supplémentaires* in French)

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Function

Function

A **function** f from a set X to a set Y assigns to each element of X exactly one element of Y .

Notation: $f : X \rightarrow Y$.

Vocabulary

Function

Function

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Notation: $f : X \rightarrow Y$.

Vocabulary

- The set X is called the **domain** of the function
- The set Y is called the **codomain** of the function.
- If the element $y \in Y$ is assigned to $x \in X$ by the function f , one says that f **maps** x to y , and this is commonly written $y = f(x)$.
- In this notation, x is the **argument** or **variable** of the function.
- A specific element x of X is a **value** of the variable, and the corresponding element of Y is the value of the function at x , or the **image** of x under f .

Remark: functional notation, arrow notation

Vocabulary

NB: In some branches of mathematics, the term map is used to mean a function.

- A **homomorphism** is a structure-preserving map between two algebraic structures of the same type (e.g. two groups, or two vector spaces).
- Homomorphisms of vector spaces are also called **linear maps**, **linear mappings** or **linear transformations**.
- A homomorphism from a vector space V to the same vector space is called an **endomorphism**.
- A bijective homomorphism is called an **isomorphism**.

ML illustration: linear maps in neural networks

A dense layer without bias and without activation is a linear map:

$$L : \mathbb{R}^d \rightarrow \mathbb{R}^k, \quad L(x) = Wx,$$

where $W \in \mathbb{R}^{k \times d}$.

A dense layer with bias is

$$x \mapsto Wx + b,$$

which is affine, but not linear unless $b = 0$.

Image (range) and Preimage (inverse image)

Image/range

The image (or range) of a function is the set of the images of all the elements in the domain. Notation: $f(X)$.

$$f(X) = \{f(x) \mid x \in X\}$$

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NB: If A is a subset of X , then the image of A under f , denoted $f(A)$, is the subset of the codomain Y consisting of all images of elements of A . We have $f(A) \subset f(X)$.

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Preimage

The **inverse image** or **preimage** under f of an element y of the codomain Y is the set of all elements of the domain X whose images under f equal y . Notation : $f^{-1}(y)$.

$$f^{-1}(y) = \{x \in X \mid f(x) = y\}$$

ML illustration: range of a linear layer

Let

$$L : \mathbb{R}^d \rightarrow \mathbb{R}^k, \quad L(x) = Wx.$$

The range of L is

$$\text{range}(L) = \{Wx \mid x \in \mathbb{R}^d\}.$$

It is the set of all outputs that the linear layer can produce.

Bijection, injection, surjection

Let $f : X \rightarrow Y$ be a function.

Injectivity

The function f is injective if $f(a) \neq f(b)$ for every two different elements a and b of X . Equivalently, f is injective iff, for every $y \in Y$, the preimage $f^{-1}(y)$ contains at most one element.

Bijection, injection, surjection

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Surjectivity

The function f is surjective if its image $f(X)$ equals its codomain Y . That is, for every element $y \in Y$, there exists an element $x \in X$ such that $f(x) = y$. Equivalently : $\forall y \in Y, f^{-1}(y) \neq \emptyset$.

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Bijection

The function f is bijective if it is both injective and surjective.

Bijection, injection, surjection

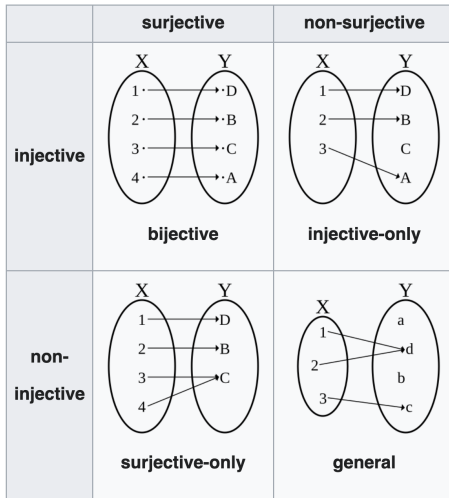


Figure 5: image source: Wikipedia

Linear transformation

Linear transformation

$L : U \rightarrow V$ is linear if for all $u_1, u_2 \in U$ and $a, b \in \mathbb{R}$,

$$L(au_1 + bu_2) = aL(u_1) + bL(u_2)$$

Example

- Show that $L : (x, y) \mapsto (x, 0)$ is linear.
- Show that $L : (x, y) \mapsto 10 \cdot (x, y)$ is linear.

Kernel / Null Space

Kernel / Null space

If $L : U \rightarrow V$ is a linear map, we define the **nullspace** (also called **kernel**^a) and denoted $\text{null}(L)$ or $\ker(L)$ of L as

$$\ker(L) = \{\mathbf{x} \in U \mid L(\mathbf{x}) = \mathbf{0}_V\}$$

^aWatch out, the word “kernel” has another meaning in machine learning.

Example

For $L(x, y) = x$, $\ker(L) = \{(0, y) \mid y \in \mathbb{R}\}$.

Reminder: the range of a linear map $L : U \rightarrow V$ writes :

$$\text{range}(L) = \{y \in V \mid \exists x \in U, L(x) = y\}.$$

Properties of subspaces and linear maps

Proposition

If $L : U \rightarrow V$ is a linear map, then

- 1 the kernel of L is a subspace of U
- 2 the range of L is a subspace of V
- 3 L is injective iff $\ker(L) = \{0_U\}$
- 4 L is surjective iff $\text{range}(L) = V$

Exercise: Proof. Hint: show that the range is closed under vector addition and scalar multiplication. Use the linearity of L to show that $L(0_U) = 0_V$.

Range of the zero map

If $L : U \rightarrow V$ is a linear map, then

$$\text{range}(L) = \{0_V\} \iff L = 0$$

Properties of subspaces and linear maps

Proposition

Let U, V be two vector spaces defined over $\mathbb{F}$, $\mathcal{B}_U = (u_1, u_2, \dots, u_n)$ a basis of U and $\mathcal{F}_V = (v_1, v_2, \dots, v_n)$ a family of vectors in V . Then there exists a unique linear map $g : U \rightarrow V$ such that

$$\forall i, 1 \leq i \leq n, g(u_i) = v_i$$

In addition:

- ❶ g is injective iff $\mathcal{F}_V$ is linearly independent (“free”).
- ❷ g is surjective iff $\mathcal{F}_V$ spans V .
- ❸ g is bijective iff $\mathcal{F}_V$ is a basis of V .

ML illustration: kernel as lost information

Let $L : \mathbb{R}^d \rightarrow \mathbb{R}^k$ be a linear map.

The kernel is

$$\ker(L) = \{x \in \mathbb{R}^d \mid L(x) = 0\}.$$

Vectors in the kernel are directions that are completely collapsed by L .

In dimensionality reduction, the kernel corresponds to information discarded by the transformation.

Example

A concrete example is PCA, where the kernel corresponds to the directions of least variance that are discarded when projecting onto the principal components. NB: PCA = Principal Component Analysis, a dimensionality reduction technique that finds a low-dimensional subspace capturing the most variance in the data (based on SVD, see Lecture on SVD).

ML illustration: PCA as compression

Let $x^{(1)}, \dots, x^{(m)} \in \mathbb{R}^d$ be centered data points:

$$\frac{1}{m} \sum_{i=1}^m x^{(i)} = 0.$$

PCA looks for orthonormal directions

$$u_1, \dots, u_k \in \mathbb{R}^d, \quad k \ll d,$$

such that the subspace

$$S = \text{span}\{u_1, \dots, u_k\}$$

captures as much variation in the data as possible. (NB: more on orthogonality in the coming lectures.)

Each data point is approximated by its projection onto S :

$$x \approx \sum_{j=1}^k (u_j^\top x) u_j.$$

Interpretation: PCA replaces a high-dimensional vector $x \in \mathbb{R}^d$ by k

ML illustration: PCA as compression

PCA gives a compressed representation of a data point.

$$x \in \mathbb{R}^d \quad \longmapsto \quad z = \begin{pmatrix} u_1^\top x \\ \vdots \\ u_k^\top x \end{pmatrix} \in \mathbb{R}^k, \quad k \ll d.$$

The vector z contains the coordinates of x in the PCA subspace.
We can reconstruct an approximation of x by

$$\hat{x} = z_1 u_1 + \cdots + z_k u_k = \sum_{j=1}^k (u_j^\top x) u_j.$$

Example: a face image with $d = 10\,000$ pixels may be represented using only $k = 100$ PCA coordinates.

Isomorphic vector spaces

Definition

We say that two vector spaces U and V are **isomorphic** if there exists an isomorphism from U to V .

NB: equivalently, we could write “from V to U ” since it is a bijection: the inverse map is also an isomorphism.

Proposition

Finite-dimensional vector spaces (defined over the same field) of the same dimension are isomorphic.

NB: Using the previous property, with $\dim(U) = \dim(V) = n$, $\mathcal{B}_U = (u_1, u_2, \dots, u_n)$ a basis of U and $\mathcal{B}_V = (v_1, v_2, \dots, v_n)$ a basis of V , we know that there exists a unique linear map $g : U \rightarrow V$ such that

$$\forall i, 1 \leq i \leq n, g(u_i) = v_i$$

and that g is bijective.

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Rank

Rank of a linear transformation

The **rank** of a linear transformation $L : U \rightarrow V$ is:

$$\text{rank}(L) = \dim(L(U))$$

where $L(U) = \{y \in V \mid y = L(x), x \in U\}$.

Reminder: $L(U)$ is called the **range** or **image** of L . We also write $\text{rank}(L) = \dim(\text{Im}(L))$.

Rank

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Reminder: $L(U)$ is called the **range** or **image** of L . We also write $\text{rank}(L) = \dim(\text{Im}(L))$.

Example

Example: $L : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $L(x, y) = x$ has rank 1.

ML illustration: rank of a data matrix

Let

$$X \in \mathbb{R}^{m \times d}$$

be a data matrix with m observations and d features.

The rank of X measures how many independent linear directions are present in the data.

- If $\text{rank}(X) < d < m$,

ML illustration: rank of a data matrix

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The rank of X measures how many independent linear directions are present in the data.

- If $\text{rank}(X) < d < m$, then some features are linearly dependent on others.
- If $m < d$, then $\text{rank}(X) \leq m$ and the data is in a low-dimensional subspace of $\mathbb{R}^d$ (and there might be less data points than features).

ML illustration: rank of a linear layer

Let $L(x) = Wx$, with $W \in \mathbb{R}^{k \times d}$.

$$\text{rank}(W) = \dim(\text{range}(L)).$$

The rank tells us the dimension of the output space that can actually be reached.

If $\text{rank}(W) < k$, then the layer cannot produce all vectors in $\mathbb{R}^k$.

Another example: LoRA (Low-Rank Adaptation) is a technique for fine-tuning large language models by adding a product of two low-rank matrices to the original weight matrices. The rank of these added matrices controls the capacity of the adaptation.

Rank-Nullity Theorem

Rank-Nullity Theorem

If U, V are finite-dimensional vector spaces and $L : U \rightarrow V$ is linear:

$$\text{rank}(L) + \dim(\ker(L)) = \dim(U)$$

Example

For $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $(x, y) \mapsto (2x, 0)$, we have $\text{rank}(L) = 1$ and $\dim(\ker(L)) = 1$.

Proof

ML illustration: rank-nullity and information loss

Let $L : U \rightarrow V$ be a linear map. The rank-nullity theorem says:

$$\text{rank}(L) + \dim(\ker(L)) = \dim(U).$$

Interpretation: the input space U is partitioned into two parts:

- $\text{rank}(L)$ measures the number of independent directions preserved.
- $\dim(\ker(L))$ measures the number of independent directions collapsed to zero.

Example: projection and information loss

Consider

$$L : \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad L(x_1, x_2, x_3) = (x_1, x_2).$$

Then

$$\text{rank}(L) = 2, \quad \ker(L) = \{(0, 0, t) \mid t \in \mathbb{R}\}.$$

Therefore

$$\dim(\ker(L)) = 1, \quad \text{rank}(L) + \dim(\ker(L)) = 2 + 1 = 3.$$

The third coordinate is discarded.

Properties of the rank

Proposition

Let $L : U \rightarrow V$ be a linear map, U and V of finite dimension.

- L is injective iff $\text{rank}(L) = \dim(U)$.
- L is surjective iff $\text{rank}(L) = \dim(V)$.
- L is bijective iff $\text{rank}(L) = \dim(U) = \dim(V)$.

Sum of subspaces

Proposition

For U and W subspaces of V ,

$$\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W)$$

Exercise: Proof.

Corollary

$\dim(U \oplus W) = \dim(U) + \dim(W)$ *for a direct sum.*

Next topics

Next class : matrices (Lecture 2)